

Andrew Collette Ph.D.

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Highly experienced scientific software developer, technical manager, and leader. In my most recent corporate role at Enthought Inc., I led the team building out our next-gen platform for analysis, and helped customers solve tough science and engineering problems with modern software tools and techniques.

BACKGROUND

Originally trained as a physicist at UCLA, I began using Python just when the science community began discovering the power of modern software techniques to accelerate research. That led me to publish a popular open-source software library for managing numerical data (h5py); in turn, this work led me to write an O'Reilly book, and finally to create my former startup company Heliosphere Research LLC. I have worked on a wide variety of projects and technologies, but mainly in engineering and engineering-management roles within the Python software ecosystem, on everything from low-level C bindings to SaaS products.

At Enthought, I had the privilege to work with customers tackling hard industrial problems by applying modern techniques, including AI/ML. Later, I took responsibility for our data analysis stack. The scientific analysis platform we built (Enthought Edge) is now used by hundreds of Enthought and customer developers, thousands of end users, and supports millions of dollars worth of business across everything from biotech to materials science.

WORK EXPERIENCE

Career Break & Professional Development

May 2025 – Present

Austin, TX and Cambridge, MA

Having spent over 15 years working as a research scientist, startup founder, and in corporate software engineering, I decided in early 2025 to focus on travel, family, and professional growth. At the end of 2025 I chose to move back to the Boston area, where I have roots. I continue to be active in software, at the moment mainly through my own projects.

Startup Exploration (Heliosphere relaunch)

August 2024 – April 2025

Austin, TX

After Enthought, I decided to take about six months or so to seriously explore re-launching my former startup, Heliosphere Research LLC. My previous work at Heliosphere focused on enabling engineers in the science and automation spaces through the Advanced Plotting Toolkit, and later a generic Python-LabVIEW bridge product that was acquired by Enthought. Areas I investigated for the re-launch included an LLM-enabled project search tool, as well as an APL-Python-CUDA bridge for running kernels written in the APL programming language directly on the GPU. Ultimately I decided not to proceed, mainly due to the levels of investment that would have been needed, and the intensely competitive market around AI products.

Vice President, Product Engineering

March 2022 – May 2024

Enthought, Austin TX

Reporting directly to the CTO, leader of a startup-style team working on Enthought's flagship scientific analysis platform *Edge*. This is in addition to overseeing maintenance work and new features for existing Enthought products. We brought Edge into a releasable state in under 12 months, and into full production in less than 24 months, on time and meeting all critical milestones. Iterated with the Product Manager on feature design, architecture, user communication, and other aspects of bringing a new product to market. Worked closely with the CTO and other company functions such as DevOps and Support, to ensure users have the best possible experience with the deployed product.

Director, Enthought Platform**January 2018 – March 2022**

Enthought, Austin TX

Managing a team of eight developers, handled the architecture and implementation for Enthought's Solution Suite (ESS). ESS is the set of core network services used by Enthought's clients and consulting developers, and includes a Python package server and extensive compiled numerical libraries for scientific data analysis. Also includes a Data Catalog service, identity management and authorization, and services for GPU-accelerated remote computing. This fully containerized stack serves as the standard technology foundation for consulting projects, allowing our specialists and customer stakeholders to focus on science and the specifics of their business problem.

Scientific Software Developer**November 2015 – January 2018**

Enthought, Austin TX

Individual contributor and later team lead for multiple Enthought consulting projects. Consulting work at Enthought is a highly agile process requiring daily coordination with customer developers and other stakeholders, as we build out hard-science applications to help them solve business problems. Key aspects of this role included working with customers to express their requirements in a clear way, and then devising a strategy to deliver rapid incremental updates for their evaluation. There was also a strong teaching element to this work; I mentored many of our early customer engineers as they began their Digital Transformation journey. I reported initially to a team lead and later to the appropriate VP-level manager responsible for customer success.

Founder**February 2013 – November 2015**

Heliosphere Research LLC, Boulder CO

Concurrent with my work at the University of Colorado, I ran a product/consulting company which served the academic and industrial communities. I worked closely with the Illinois-based nonprofit HDF Group, for whom I provided Python training. Heliosphere's flagship product was the Advanced Plotting Toolkit, a set of LabVIEW bindings for the "matplotlib" visualization library. This product was sold for approximately two years on National Instruments' LabVIEW Tools Network. I later licensed the code to Enthought, where it formed the foundation of Enthought's product Python Integration Toolkit for LabVIEW.

Research Scientist**May 2013 – November 2015**

Laboratory for Atmospheric and Space Physics, Univ. Colorado Boulder

Performed NASA-funded laboratory research into the effects of high-speed impacts on spacecraft and planetary surfaces. Experiments focused on in-situ detection of dust grains by spacecraft equipped with antenna systems. Research was carried out at the CU Boulder IMPACT accelerator facility, which I helped construct. Regular publication in major journals, proposals to NASA and NSF, and close collaboration with others in the field. Daily management of graduate and undergraduate students on lab projects.

Postdoctoral Researcher**July 2010 – May 2013**

Laboratory for Atmospheric and Space Physics, Univ. Colorado Boulder

Took a leading role in the construction and commissioning of the new hypervelocity dust-grain accelerator at CU's Institute for Modeling Plasma, Atmospheres and Cosmic Dust (IMPACT). Performed and published many of the first results from this facility, on impact-produced light flash and neutral gas generation.

EDUCATION

University of California, Los Angeles (Los Angeles, CA)

Ph.D., Physics – 2010. Dissertation on the physics of expanding laser-produced plasmas.

University of Rochester (Rochester, NY)

B.S., Physics – 2004.

SKILLS & INDUSTRY EXPERIENCE

Industries: While at Enthought I worked with customers across a wide variety of industries: innovative DNA sequencing; fine chemicals, including development of new photoresist products for semiconductor production; development of synthetic rubber; seismic data analysis for oil and gas exploration; well treatment and decline curves for oil production; and many more. As an academic and at Heliosphere I worked with partners and customers in industrial automation; weather forecasting; NASA-funded projects for lunar exploration and ISRU; and others related to management of large volumes of numerical data.

Languages: Python, including low-level Cython integration and package building. Extensive C and Cython experience, mainly in integrating third-party libraries with CPython. LabVIEW for integration and laboratory automation. Recent others include Javascript and TypeScript; languages I've used in the past include Go, Java, C++, and IDL (by NV5 Geospatial).

Cloud Services: Very familiar with AWS and Kubernetes as this is our standard platform at Enthought; emphasis on S3, EC2, EKS, IAM, and the Cost Explorer API. I also have some Azure experience as it was the standard platform for many customers.

Other Tools and Frameworks: Container technologies including local Kubernetes, Docker/DockerHub/Quay.io; scientific Python stack including NumPy/SciPy, FastAPI, Flask, PostgreSQL, Linux, and enough React/HTML/CSS to be dangerous.

Management: "Enthought Agile," a technique for rapidly and responsively developing software in close coordination with technical stakeholders. Continuous 1:1 mentoring and supervisory responsibility for the developers on my team. Management tools involve ZenHub (Kanban/Agile), JIRA, Redmine, or just plain GitHub Issues and Projects.

SELECTED PUBLICATIONS

A. Collette. *Python and HDF5: Unlocking Scientific Data*. O'Reilly, 2013

A. Collette, G. Meyer, D. Malaspina, Z. Sternovsky. *Laboratory investigation of antenna signals from dust impacts on spacecraft*. JGR Space Physics 120, 2015, doi:10.1002/2015JA021198.

A. Collette, E. Gruen, D. Malaspina, Z. Sternovsky. *Micrometeoroid impact charge yield for common spacecraft materials*. JGR Space Physics 119, 6019-6026, 2014, doi:10.1002/2014JA020042.

A. Collette, Z. Sternovsky, M. Horanyi. *Production of neutral gas by micrometeoroid impacts*. Icarus 227, 89-93, 2014.

A. Collette, K. Drake, A. Mocker, Z. Sternovsky, T. Munsat, M. Horanyi. *Time-resolved temperature measurements in hypervelocity dust impact*. Planetary and Space Science 89, 58-62, 2013.

A. Shu, **A. Collette** et al. *3 MV hypervelocity dust accelerator at the Colorado Center for Lunar Dust and Atmospheric Studies*. Review of Scientific Instruments 83, 075108, 2012.

A. Collette and S. Robertson. *An ion analyzer for the lunar surface with E-parallel-B*. Advances in Space Research 50, 1592-1599, 2012.

A. Collette and W. Gekelman. *Structure of an Exploding Laser-Produced Plasma*. Physical Review Letters 105, 195003, 2010.